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HNG INTERNSHIP STAGE 2

TRADEZONE DATA ANALYSIS REPORT (PART C)

TOOL USED: POSTGRESQL (pgAdmin 4)

## 1. INTRODUCTION

### 1.1 Business Context

TradeZone is a rapidly expanding Nigerian e-commerce platform that links buyers and sellers in Lagos, Abuja, Kano, Port Harcourt and Ibadan. The platform has been active over two full years (20232024) and the information gathered throughout this time is the basis of this analysis.

The Head of Growth and the Head of Seller Operations sought a joint data analysis, before the 2025 planning cycle. This report shows the results of eight business questions that were answered with SQL with the assistance of a data cleaning exercise that preceded analysis.

### 1.2 Dataset Overview

| S/N | Attribute       | Detail                       |
|-----|-----------------|------------------------------|
| 1.  | Database Name   | tradezone                    |
| 2.  | Analysis Period | January 2023 – December 2024 |
| 3.  | Tool Used       | PostgreSQL via pgAdmin 4     |
| 4.  | Total Tables    | 7                            |

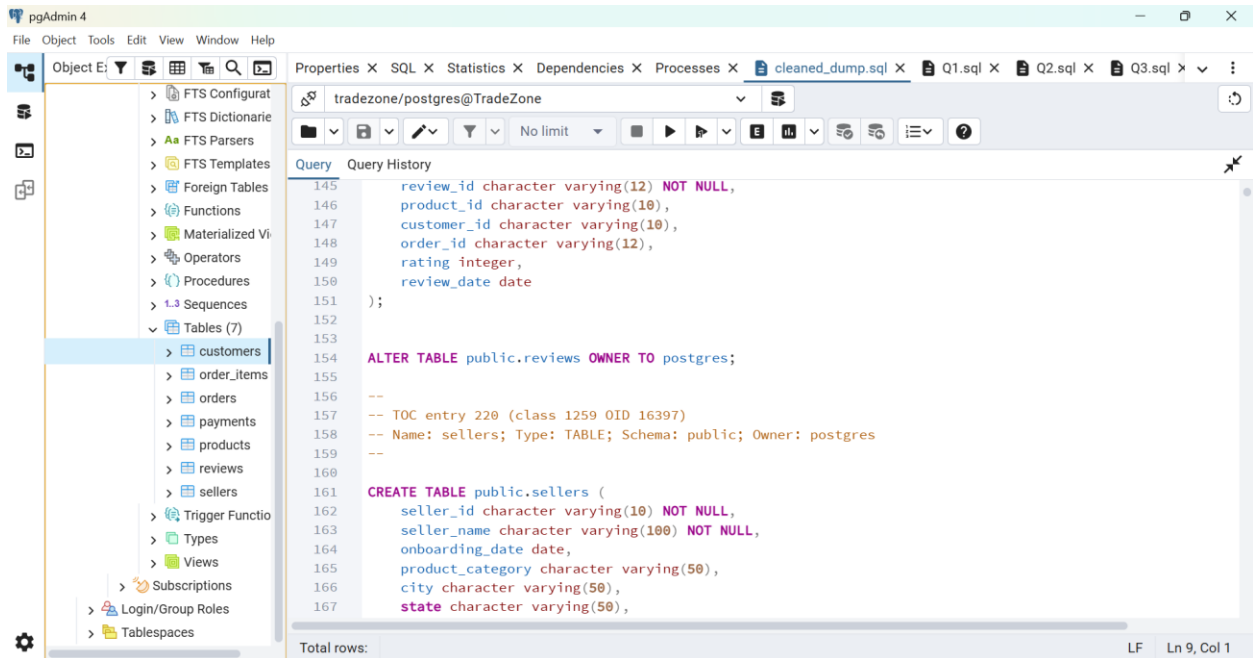


Fig 1.1: pgAdmin showing all 7 tables

### 1.3 Table Descriptions

| S/N | Attribute   | Description                                                |
|-----|-------------|------------------------------------------------------------|
| 1.  | customers   | 850 customer records with location and signup information. |
| 2.  | sellers     | seller profiles with category and location.                |
| 3.  | products    | product catalogue with pricing.                            |
| 4.  | orders      | all transactions placed on the platform.                   |
| 5.  | order_items | line items within each order.                              |
| 6.  | payments    | payment method and amount per order.                       |
| 7.  | reviews     | customer ratings per product.                              |

## **2. DATA CLEANING & PREPARATION**

The dataset was cleaned in order to analyze, make it accurate and consistent. All cleaning was performed using SQL in PostgreSQL.

### **2.1 City Name Standardization**

City names contained inconsistent formatting which include all-caps entries (LAGOS, ABUJA, IBADAN), trailing spaces (Kano, Lagos ), and alternate spellings of Port Harcourt (Port-Harcourt, Port Harcourt).

The INITCAP () and TRIM () functions have been used on both the customers and sellers' tables to streamline all the names of the cities to Title Case without any additional spaces. Port Harcourt variations were fixed with the help of a WHERE clause which was aimed at the known incorrect spellings.

### **2.2 Product Category Normalization**

The products and sellers tables had their product category names normalized to Title Case with INITCAP (TRIM (category)) to make sure that they are grouped the same way in analysis queries.

### **2.3 Duplicate Records**

A check for duplicate email addresses in the customers table was conducted. Where duplicates were found, records could not be safely deleted because customer IDs were referenced in the orders table (foreign key constraint). These duplicates were flagged and documented but retained to preserve order history integrity.

### **2.4 Invalid Review Ratings**

A review rating of 0 was found in record REV00718. Since valid ratings must fall between 1 and 5, this record was deleted. A rating of 0 cannot represent a genuine customer experience on a 1–5 scale and would distort average rating calculations if retained.

### **2.5 NULL Product Prices**

Four products were found to have NULL unit prices:

1. PROD0088: Kitchen Knife Set 7 Piece.
2. PROD0104: Standing Fan 18 inch 5-Speed.
3. PROD0205: Ribena Blackcurrant Drink 1L
4. PROD0245: JAMB CBT Practice Questions

These records were retained but flagged. Their NULL prices caused revenue to appear as NULL in the product performance query (Q2).

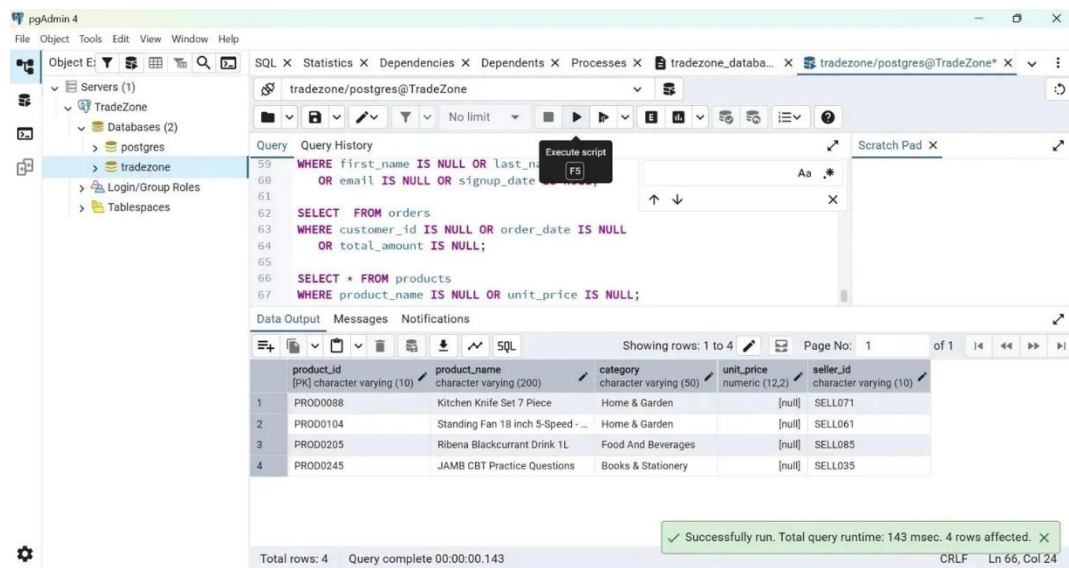


fig 2: Cleaning script Messages tab

## 2.6 Order Amount Validation

The total of the amount in each order was compared with the amount of the line items of the order in order items. Orders with a difference of over 10 were put under review. This confirms whether recorded totals are consistent with item-level data.

## 2.7 NULL Value Checks

Critical columns in customers, orders and products were checked for NULL values to identify any records that would cause analysis errors.

### 3. BUSINESS QUESTIONS & QUERY RESULTS

#### 3.1 Question 1: Customer Acquisition & 30-Day Conversion

Business Question: Which are the top 5 states by new customer sign-ups in 2024, and what percentage of these new customers made at least one purchase within their first 30 days?

Approach: A CTE (Common Table Expression) was used to first isolate customers who signed up in 2024, then join to orders to check if each customer placed an order within 30 days of signing up. A LEFT JOIN ensured customers with no orders were still counted.

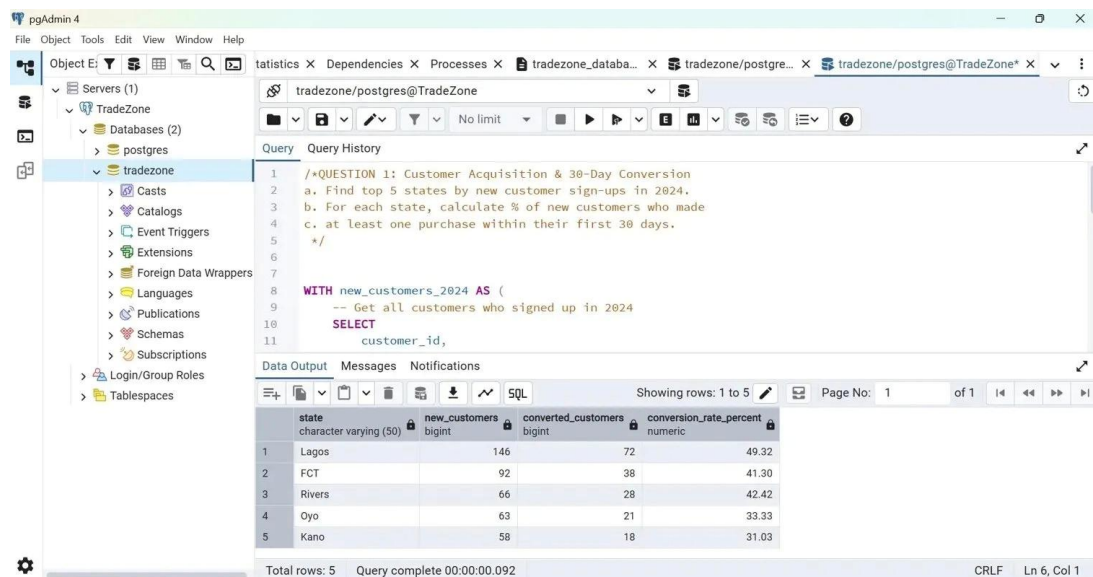


fig 3.1: Q1 results

Results:

| State  | New Customers | Converted | Rate   |
|--------|---------------|-----------|--------|
| Lagos  | 146           | 72        | 49.32% |
| FCT    | 92            | 38        | 41.30% |
| Rivers | 66            | 28        | 42.42% |
| Oyo    | 63            | 21        | 33.33% |
| Kano   | 58            | 18        | 31.03% |

Insight: Lagos leads in acquisition with 146 new customers but only converts 49.32% within 30 days. Kano has the lowest conversion rate at 31.03%. No state achieved 50% 30-day conversion, indicating a platform-wide onboarding problem.

### 3.2 Question 2: Product Performance

Business Question: What are the top 10 products by total revenue in 2024?

Approach: order\_items was joined to orders and products, filtered to 2024, and revenue was summed per product using SUM(line\_total).

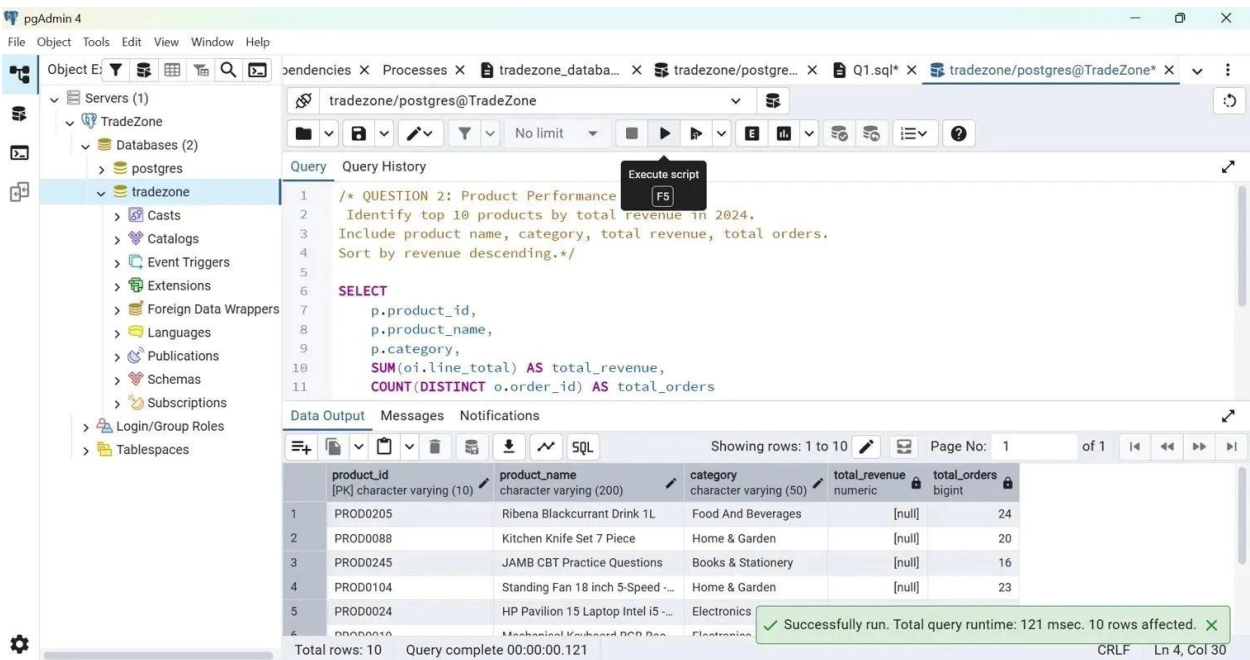


fig 3.2: Q2 results

Insight: Four of the top-ranked products showed NULL revenue due to missing unit prices identified in the cleaning phase. These products ranked highly by order count, suggesting strong demand despite missing price data. This is a data quality gap that requires urgent attention from the product team.

### 3.3 Question 3: Seller Fulfilment Efficiency

Business Question: Which top 20 sellers have the fastest average fulfilment time among those with at least 20 completed orders?

Approach: Delivery time in hours was calculated using `EXTRACT (EPOCH FROM (delivery_date - order_date)) / 3600`. Only Delivered orders were included. Results were filtered using `HAVING COUNT >= 20`.

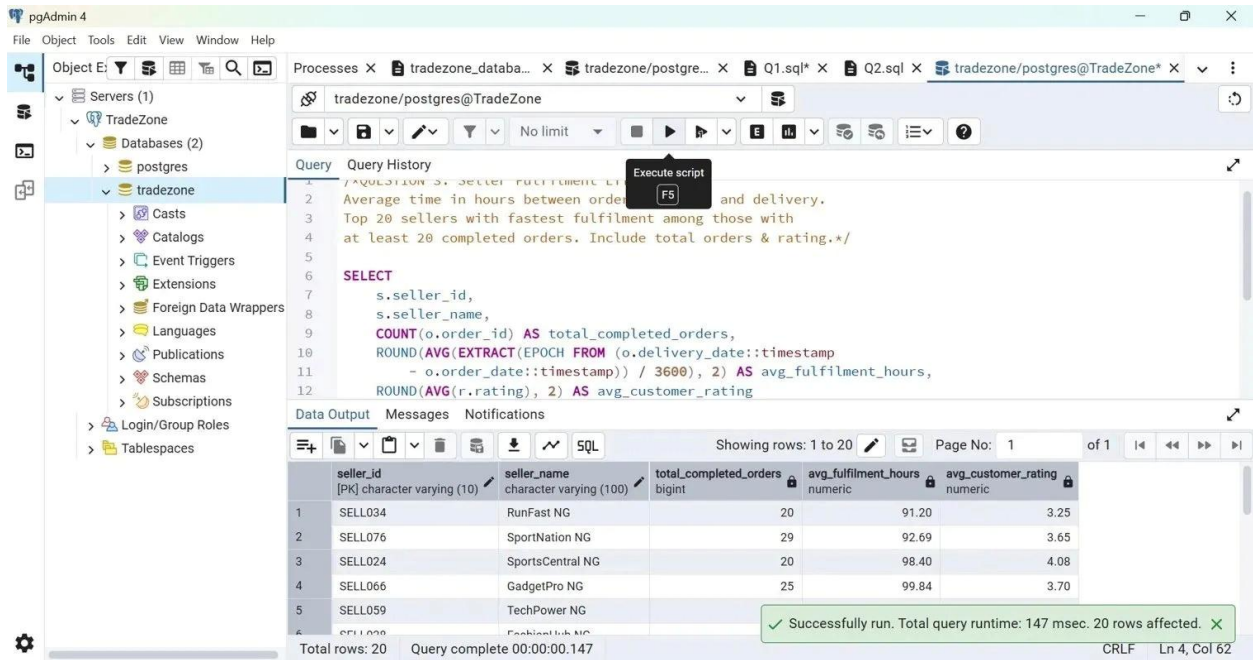


fig 3.3: Q3 results

Results (Top 5):

| S/N | Seller         | Orders | Avg Hours | Avg Rating |
|-----|----------------|--------|-----------|------------|
| 1.  | RunFast NG     | 20     | 91.20     | 3.25       |
| 2.  | SportNation NG | 29     | 92.69     | 3.65       |
| 3.  | GadgetPro NG   | 25     | 99.84     | 3.70       |
| 4.  | TechPower NG   | (NULL) | (NULL)    | (NULL)     |

Insight: The fastest seller (RunFast NG) has the lowest rating among the top 5. SportsCentral NG is slightly slower but rated significantly higher at 4.08. Speed alone does not determine customer satisfaction.

### 3.4 Question 4: Quarterly Revenue Trends

Business Question: How does quarterly revenue compare across 2023 and 2024?

Approach: EXTRACT(YEAR) and EXTRACT(QUARTER) were used to group orders by quarter. Cancelled orders were excluded.

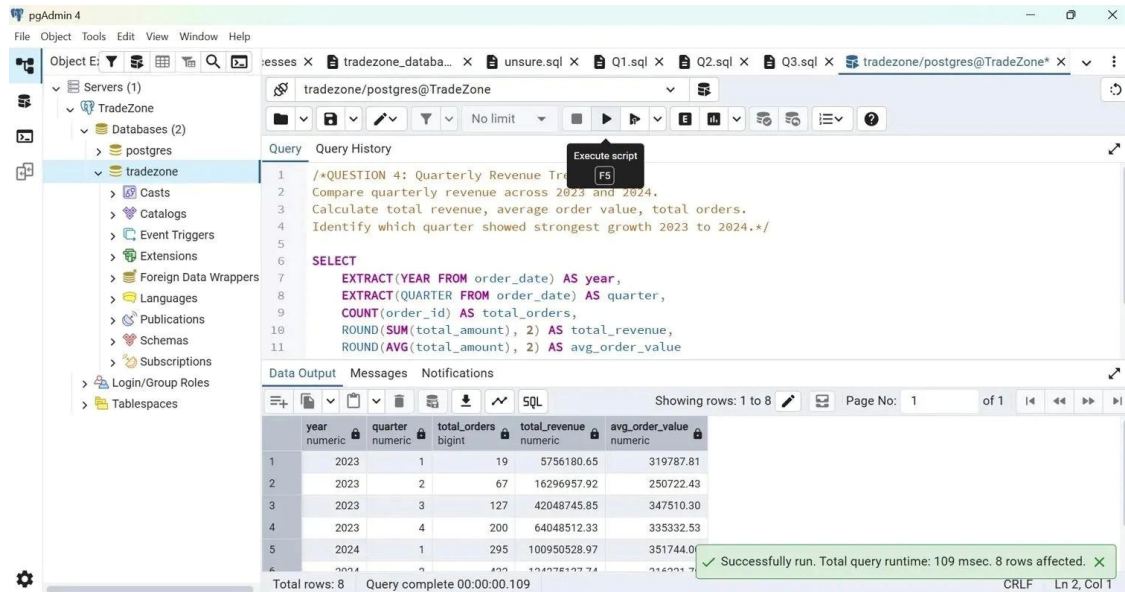


Fig 3.4: Q4 results

Insight: Q1 2024 (¥100.9M) vs Q1 2023 (¥5.7M) represents the strongest year-on-year quarterly growth a 1,653% increase. The platform grew dramatically in 2024 across all quarters.

### 3.5 Question 5: Customer Spend Segmentation

Business Question: How are customers distributed across spending segments in 2024?

Approach: Total spend per customer was calculated using SUM(total\_amount), then grouped using a CASE statement into High, Medium and Low Spender categories.

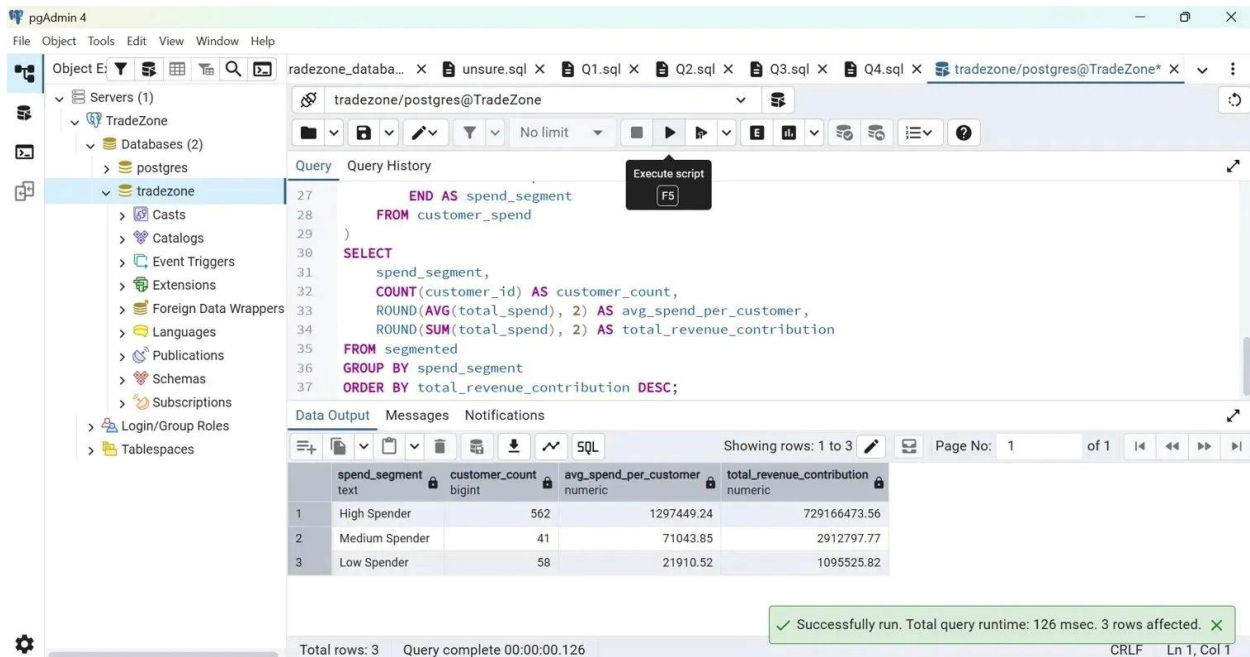


fig 3.5: Q5 results

Insight: 562 High Spenders generate ₦729M approximately 98% of total 2024 revenue. The business is critically dependent on this group. Loss of even 10% of high spenders would cost over ₦72M in revenue.

### 3.6 Question 6: Payment Method Preferences by State

Business Question: What payment methods are preferred in each state?

Approach: Payments were joined to orders and customers, grouped by state and payment method. A RANK () window function identified the most popular method per state.

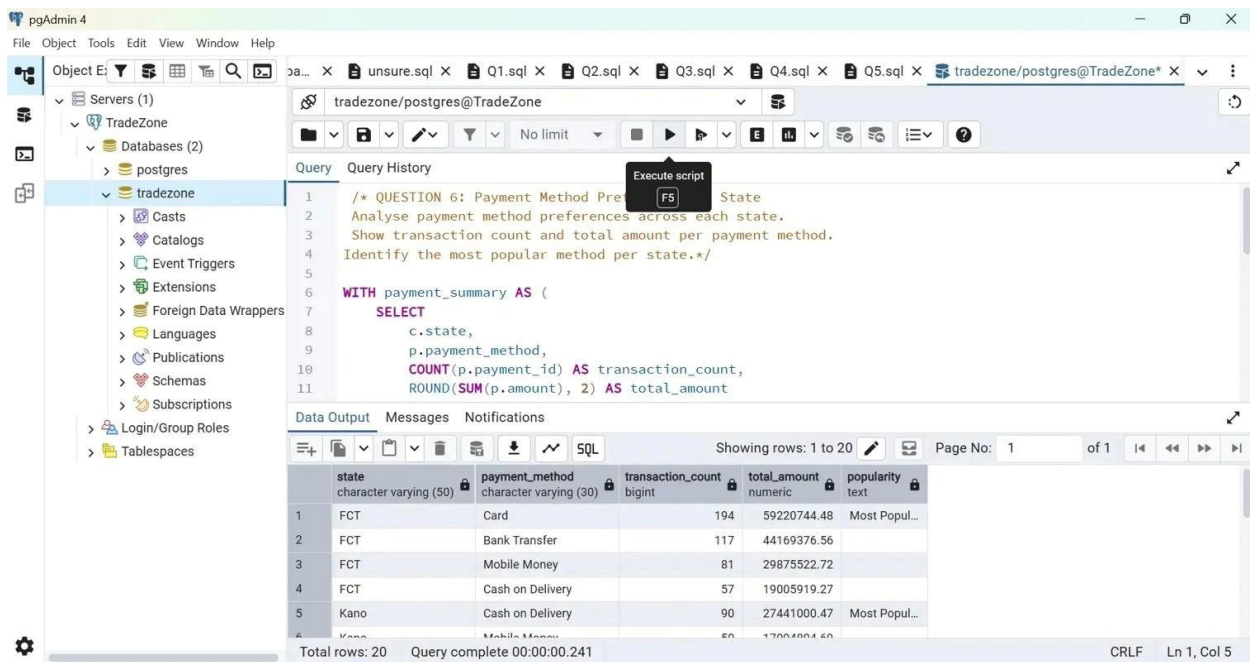


Fig 3.6: Q6 results

Results:

| State  | Most Popular Method | Transactions |
|--------|---------------------|--------------|
| FCT    | Card                | 194          |
| Kano   | Cash on Delivery    | 90           |
| Lagos  | Card                | —            |
| Oyo    | Card                | —            |
| Rivers | Card                | —            |

Insight: Card payments dominate in FCT, Lagos, Oyo and Rivers indicating digital payment comfort in southern and central states. Kano strongly prefers Cash on Delivery, reflecting different trust levels with digital payments in the north. Payment infrastructure strategy should be state-specific.

### 3.7 Question 7: Review Ratings and Sales Performance

Business Question: How do products grouped by rating category compare in revenue and pricing?

Approach: Average ratings per product were calculated and grouped into High, Mid and Low Rated categories using a CASE statement. Revenue was summed from order\_items.

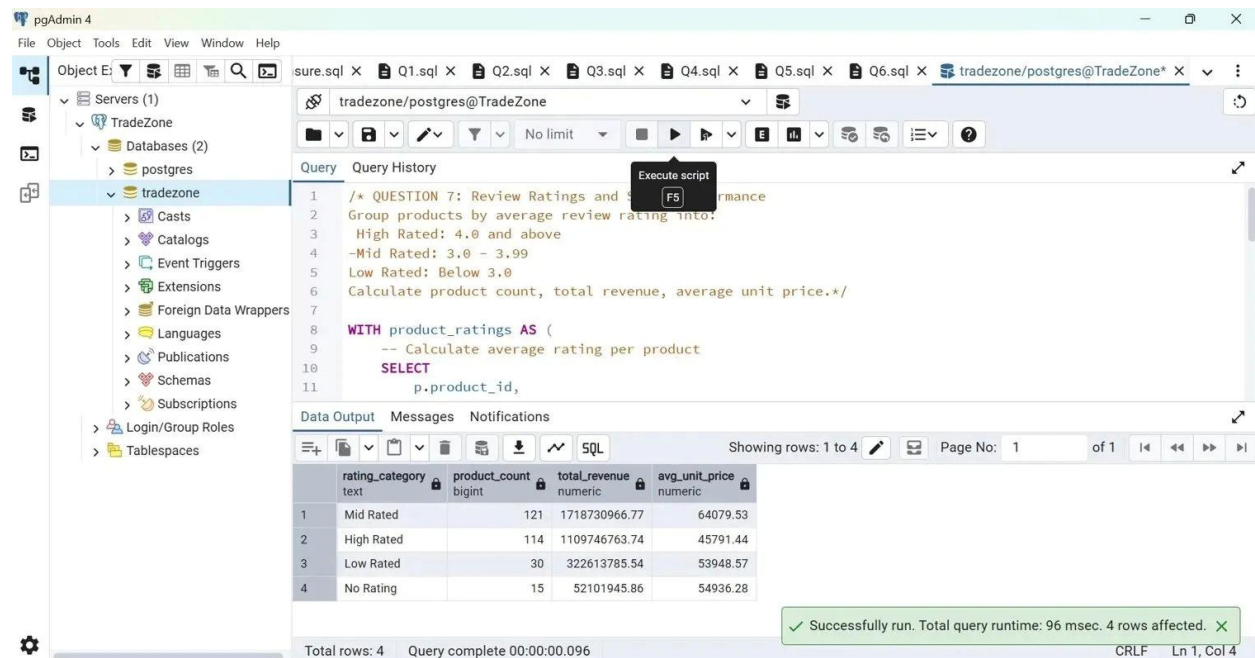


Fig3.7: Q7 results

Insight: Mid Rated products (3.0–3.99) generate more revenue than High Rated products. This suggests customers prioritize value and availability over perfect ratings. Low Rated products still generate \$322M, indicating that poor reviews do not necessarily stop purchases but they may affect long-term retention.

### 3.8 Question 8: Top Seller Bonus Qualification

Business Question: Which top 10 sellers in 2024 qualify for a bonus based on revenue, order count and rating?

Approach: Sellers were filtered to those with at least 10 completed orders and an average rating of 4.0 or above, then ranked by total revenue.

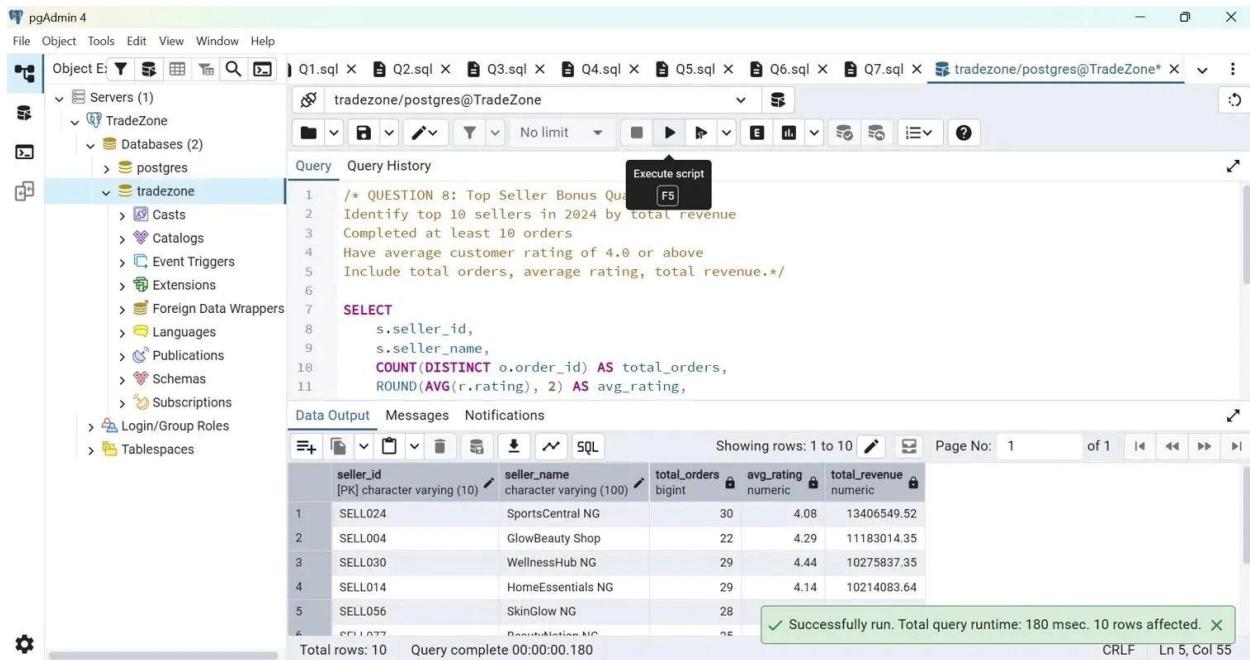


Fig3.8: Q8 results

Results (Top 5):

| S/N | Seller            | Orders | Avg Rating | Revenue     |
|-----|-------------------|--------|------------|-------------|
| 1   | SportsCentral NG  | 30     | 4.08       | ₦13,406,550 |
| 2   | GlowBeauty Shop   | 22     | 4.29       | ₦11,183,014 |
| 3   | WellnessHub NG    | 29     | 4.44       | ₦10,275,837 |
| 4   | HomeEssentials NG | 29     | 4.14       | ₦10,214,084 |
| 5   | SkinGlow NG       | 28     | 4.22       | ₦9,889,042  |

Insight: These 10 sellers combine high revenue with strong customer satisfaction. They represent TradeZone's most valuable seller partners and should be prioritised for retention and growth support.

## **4. EXECUTIVE SUMMARY**

TradeZone experienced explosive revenue growth between 2023 and 2024, with Q1 2024 revenue of ₦100.9M compared to Q1 2023's ₦5.7M a 1,653% increase. However this growth masks critical vulnerabilities. Customer conversion rates remain below 50% across all states, and platform revenue is dangerously concentrated in 562 high spenders contributing ₦729M. Without deliberate retention and diversification strategies, this growth remains fragile.

### **4.1. KEY FINDINGS**

**1. Customer Conversion Is Critically Low (Q1).**

None of the states had a 50% 30-day conversion in 2024. Lagos led with 49.32% while Kano was lowest at 31.03%. More than half of the new sign-ups do not make a first purchase. This is a significant lost revenue potential and shows that the onboarding experience is not doing a good job at motivating new customers to purchase.

**2. Revenue Is Dangerously Concentrated (Q5)**

562 High Spenders brought 729M - about 98% of total 2024 platform revenues. There were only 99 customers who were of Medium or Low spend. In case a large percentage of the heavy spenders churn, the platform is in a dire revenue crisis with no cushion.

**3. Fast Fulfilment Does Not Guarantee High Ratings (Q3)**

Run Fast NG has the longest time of 91.20 hours but with a low rating of 3.25. SportsCentral NG has slightly lower 98.40 hours but was rated 4.08. Speed is not important to the customers; product quality, packaging and communication are as important or more.

## 5. RECOMMENDATIONS

To improve customer conversion and retention, this is what I recommend.

The first is to launch a “30-day new customer activation campaign”. This should focus on guiding new users from sign-up to their first purchase through a simple but structured onboarding journey. For example, new customers can receive a first-purchase discount voucher within 48 hours of signing up, followed by a personalized product recommendation email around Day 7, and a reminder nudge closer to Day 25.

This initiative would be led by the Head of Growth, specifically across CRM and email marketing. If executed well, it is expected that within 60 to 90 days, the 30-day conversion rate could increase from the current average of 39% to at least 55% across all states.

The second recommendation is to introduce a “high spender loyalty and retention programme”. This would involve identifying the 562 top-spending customers and enrolling them in a VIP experience. Benefits could include priority delivery, access to exclusive sellers, and quarterly loyalty rewards to keep them engaged.

In addition, a churn monitoring dashboard should be created specifically for this group to track behaviour and proactively manage retention risks. This effort would be jointly handled by the Head of Seller Operations (for delivery improvements) and the Head of Growth (for loyalty programme design). The goal is to reduce churn among high spenders by at least 15% within 90 days, helping to protect the ₦729M revenue they currently generate.

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### 5.1.DATA QUALITY NOTES

While reviewing the data, a few minor issues came up that are worth noting, although they don’t significantly affect the overall analysis.

First, four products (PROD0088, PROD0104, PROD0205, and PROD0245) didn’t have unit prices recorded. Because of this, their revenue showed up as NULL in Query 2. These records were kept but flagged, as removing them could hide potentially important information. If any of

these products are high-value, it means total revenue and product rankings may currently be slightly understated. Once the correct prices are added, the results could shift.

Secondly, one review (REV00718) had a rating of 0, which falls outside the normal 1–5 scale. This entry was removed to keep the dataset consistent. Since there are still 817 valid reviews, this has little impact overall. However, if the rating was intentionally recorded that way, removing it may slightly increase the average rating of the affected product. These issues are small and don't change the main insights, but they've been highlighted for transparency and to point out where the data can be improved.

## **5.2. WHAT THE DATA CANNOT TELL US**

The existing data is unable to respond to: Why are customers not coming back after purchasing the first time? Orders, reviews and customer tables are what has occurred and not why. No information about post-purchase surveys, requests of returns/refunds, communication with the customer service, or platform session activity. More information was required: post purchase feedback surveys, customer service ticket logs and Web clickstream data of where people abandon the purchase process. This would enable the Growth team to know whether it is low conversion problem of a product, price, trust or experience.